ATLAS Software Tables

Device Table, Sample Rate Plan, Communication Map, and Pinout Sheet

Purpose

This document gives insight into the hardware interface list for ATLAS. ATLAS stands for Avionics Telemetry and Logging Acquisition System.

ATLAS is hopefully going to be the secondary flight computer, telemetry system, data logger, recovery controller, and sensor acquisition system for Monarch and future rockets.

1. Device Table and Sample Rate Plan

Device	Part number	Purpose	Interface and bus	Main data	Target rates (TBD)	Log plan	Notes
Main MCU	STM32F407VGT6	Runs firmware, state machine, logging, telemetry, recovery logic, and safety logic.	All interfaces	All system data	Safety loop: 500 Hz to 1 kHz. Health loop: 100 Hz to 200 Hz.	N/A	Use watchdogs, fault codes, safe boot defaults, and clear startup checks.
IMU 1	LSM6DS3TR-C	Main motion sensor for acceleration and gyro data.	SPI, SPI1	ax, ay, az, gx, gy, gz, temp	Wait: 100 Hz to 200 Hz. Flight: 400 Hz to 800 Hz. Landed: off or 25 Hz.	Log raw and filtered motion data during flight.	Use FIFO or data-ready interrupt. Keep exact ODR in firmware config.
IMU 2	BMI270	Second motion sensor for fusion, backup motion data, and drift checking.	SPI, SPI1	ax, ay, az, gx, gy, gz, temp	Wait: 100 Hz to 200 Hz. Flight: 400 Hz to 800 Hz. Landed: off or 25 Hz.	Log raw and filtered motion data during flight.	Use as cross-check against IMU 1.
High-g accelerometer	ADXL375BCCZ	Detect high-g launch events and record motor boost loads.	SPI, SPI1	ax, ay, az	Wait: 100 Hz. Flight: 800 Hz to 1 kHz. Landed: off.	Log raw high-g data during boost and coast.	Exact supported ODR still needs ADXL375 datasheet in the repo.
Barometer	MS5607	Altitude, pressure, temperature, apogee support, and descent tracking.	I2C planned, I2C1	pressure, temp, altitude estimate	Wait: 10 Hz to 25 Hz. Flight: 50 Hz to 100 Hz. Landed: 1 Hz to 5 Hz.	Log pressure, temperature, and altitude.	Choose OSR based on noise and update rate. Higher OSR gives slower conversion.
Magnetometer	LIS3MDL	Heading reference, yaw drift correction support, and post-flight orientation review.	I2C planned, I2C1	mx, my, mz	Wait: 10 Hz. Flight: 40 Hz to 80 Hz. Landed: off or 1 Hz.	Log during flight for analysis.	Not a pyro safety sensor. Use carefully near magnetic noise.
Power monitor	INA260	Battery power health, current draw, and power fault detection.	I2C, I2C1	bus voltage, current, power	Wait: 5 Hz to 10 Hz. Flight: 20 Hz. Landed: 1 Hz.	Log power health.	Use alert pin.
GNSS receiver	MAX-M10S	Position, speed, altitude, time, and recovery location.	UART	lat, lon, altitude, speed, fix, satellites, time	Wait: 1 Hz to 5 Hz. Flight: 5 Hz to 10 Hz. Landed: 1 Hz.	Log GNSS when fix is valid.	Handoff should lock this as UART before coding starts.
LoRa radio	RFM95W-915	Sends telemetry from rocket to ground station.	SPI plus DIO pins, SPI3	telemetry packets, RSSI, SNR, status	Wait: 1 Hz. Flight: 5 Hz to 10 Hz. Landed: 1 Hz.	Log packet count and link status.	Keep payload small. Do not block the flight loop while transmitting.
Flash memory	W25Q128JVS	Black box backup storage and fault or event history.	SPI, SPI3	event records, snapshots, config backup	Event based. Flight snapshots: 10 Hz to 50 Hz.	Store compact binary records.	Do not erase during critical flight events. Use a ring buffer or page-based log.
microSD card	microSD	Main removable flight log storage.	SDIO	CSV flight log	Wait: 10 Hz to 25 Hz output. Flight: 50 Hz to 200 Hz output. Landed: 1 Hz.	Main log file.	Use RAM buffers. Never let SD writes block pyro logic.
Drogue ejection circuit	Drogue pyro	Fires drogue deployment charge.	GPIO plus ADC	fire command, continuity, sense	Event driven. Health check: 5 Hz to 20 Hz.	Log continuity and fire event.	Safety critical. Output must default OFF during boot, reset, fault, and crash.
Main ejection circuit	Main pyro	Fires main deployment charge.	GPIO plus ADC	fire command, continuity, sense	Event driven. Health check: 5 Hz to 20 Hz.	Log continuity and fire event.	Safety critical. Output must default OFF during boot, reset, fault, and crash.
Camera control	Camera power control	Controls 5 V camera power after landing or command event.	GPIO	enable state	Event driven.	Log camera power state.	Keep camera logic separate from recovery safety logic.
Buzzer	Buzzer circuit	Beep codes for boot, arm, fault, and recovery.	GPIO or PWM	state feedback	Event driven.	Log fault source, not every beep.	Must be non-blocking.
LEDs	Status LEDs	Shows power, arm, fault, GPS, LoRa, SD, and boot status.	GPIO	state feedback	Event driven.	No continuous LED log needed.	Must be non-blocking.

Recommended Firmware Loop Rates

Loop	Purpose	Suggested rate	Runs in states	Notes
Safety loop	Pyro lockout, arming state, fault state, and watchdog feed rules.	500 Hz to 1 kHz	All states	Must run even during SD or LoRa activity.
Sensor acquisition loop	Reads IMUs, high-g accel, barometer, power monitor, and magnetometer.	100 Hz to 1 kHz	Wait for launch, ascent, descent	Use device-ready flags where possible.
State estimation loop	Calculates altitude, velocity, attitude estimates, launch detect, and apogee detect.	100 Hz to 500 Hz	Wait for launch, ascent, descent	Use filtered data, not raw single samples.
Logging loop	Packs data and writes to RAM buffer.	50 Hz to 200 Hz	All flight states	SD task writes buffer later.
SD write task	Flushes RAM buffer to microSD.	As needed	All states except critical one-shot events	Must not block recovery logic.
Flash black box task	Writes event records and backup snapshots.	Event based, 10 Hz to 50 Hz during flight	All states	Keep records compact.
Telemetry task	Sends LoRa packets.	1 Hz to 10 Hz	All states	Lower rate after landing.
GPS task	GNSS messages.	Same as GPS update rate	Wait for launch, flight, landed	Use non-blocking receive.
User feedback task	LEDs and buzzer.	1 Hz to 20 Hz	All states	Non-blocking only.

Recommended Logged Data Rates

Data group	Suggested log rate	Reason
Raw IMU data	100 Hz to 500 Hz	Useful post-flight graphs without huge files.
High-g accel during boost	500 Hz to 1 kHz	Captures motor boost and vibration better.
Barometer altitude	50 Hz	Good for ascent, apogee, and descent trend.
GPS	5 Hz to 10 Hz when valid	Limited by GNSS update rate.
Power monitor	10 Hz to 20 Hz	Battery data changes slower than motion data.
Magnetometer	25 Hz to 50 Hz	Useful for orientation review, not main recovery logic.
State and event records	Every event plus 10 Hz heartbeat	Helps debug flight logic.
Telemetry status	5 Hz to 10 Hz in flight, 1 Hz landed	Keeps packet size and airtime under control.

2. Communication Map and Pinout Sheet

Communication Map

Bus or group	MCU peripheral	Known or recommended MCU pins	Connected devices	Signals	Speed target	Notes
Main sensor SPI	SPI1	PA5 SCK, PA6 MISO, PA7 MOSI	LSM6DS3TR-C, BMI270, ADXL375	SCK, MISO, MOSI, CS per device, INT per sensor	Start 5 MHz, increase after testing	Shared high-speed motion bus. Every device needs its own CS.
Flash and LoRa SPI	SPI3	PB3 SCK, PB4 MISO, PB5 MOSI	W25Q128JVS, RFM95W-915	SCK, MISO, MOSI, CS_FLASH, CS_LORA, RFM_DIO pins, RFM_RESET	Start 2 MHz to 8 MHz	LoRa timing must not block flash writes or flight logic.
microSD	SDIO	PC8 D0, PC9 D1, PC10 D2, PC11 D3, PC12 CK, PD2 CMD	microSD socket	D0, D1, D2, D3, CK, CMD, card detect if routed	Start 1-bit low speed, then 4-bit SDIO	Use buffered writes. Keep pyro logic independent from SD activity.
Sensor I2C	I2C1	PB6 SCL, PB7 SDA	MS5607, LIS3MDL, INA260	SCL, SDA, optional alert or data-ready pins	400 kHz	Confirm pullups and device addresses before full firmware work.
GPS interface	UART	PA0 TX, PA1 RX	MAX-M10S	TX, RX, TIMEPULSE, NRESET test pad	UART 9600 to 115200	Lock this as UART before coding starts.
USB	USB OTG FS	PA11 USB_DM, PA12 USB_DP	USB-C connector	D-, D+	USB FS	Used for programming, config, and possible debug.
Debug	SWD	PA13 SWDIO, PA14 SWCLK, NRST	ST-Link	SWDIO, SWCLK, NRST, 3V3, GND	SWD	Must stay accessible on every board revision.
Pyro outputs	GPIO	Main Arm/Sense PD15, Main Fire PD14, Apo Arm/Sense PD10, Apo Fire PD11	Drogue and main ejection circuits	MAIN_FIRE, DROGUE_FIRE, PYRO_ARM, continuity and sense lines	Event driven	Highest safety review priority. Default OFF at boot.

Analog sense	ADC GPIO	Main Sense PD13, Apo Sense PD9, VBatt sense PC2	Battery sense, pyro sense, continuity sense	ADC inputs	10 Hz to 100 Hz depending signal	Scale to engineering units in one module.
Camera control	GPIO	PE2	Camera power circuit	CAM_PWR_DISABLE	Event driven	Startup state low.
User feedback	GPIO or PWM	Buzzer PD3, LEDs PE1 & PB12	Buzzer and LEDs	BUZZER, LED_STATUS, LED_GPS, LED_LORA, LED_SD, LED_FAULT	Event driven	Non-blocking patterns only.
Boot and reset	GPIO and hardware pins	BOOT0, NRST	Boot switch, reset circuit	BOOT0, NRST	Hardware controlled	BOOT0 default should be normal flash boot.

Pinout Sheet

Use this as the first software pin table

Net name	MCU pin	MCU mode	Peripheral or circuit
SPI1_SCK	PA5 (30)	AF SPI1	IMU 1, IMU 2, high-g accel
SPI1_MISO	PA6 (31)	AF SPI1	IMU 1, IMU 2, high-g accel
SPI1_MOSI	PA7 (32)	AF SPI1	IMU 1, IMU 2, high-g accel
CS_IMU1	PE12 (43)	GPIO output	LSM6DS3TR-C
CS_IMU2	PE8 (39)	GPIO output	BMI270
CS_HIGHG	PB2 (37)	GPIO output	ADXL375BCCZ
INT_IMU1	PE11 (42)	GPIO input EXTI	LSM6DS3TR-C interrupt
INT_IMU2	PE9 (40)	GPIO input EXTI	BMI270 interrupt
INT_HIGHG	PB1 (36)	GPIO input EXTI	ADXL375 interrupt
I2C1_SCL	PB6 (92)	AF I2C1	INA260
I2C1_SDA	PB7 (93)	AF I2C1	INA260
INA260_ALERT	PB9 (96)	GPIO input EXTI	INA260 alert
I2C2_SCL	PB10 (47)	AF I2C1	MS5607, LIS3MDL
I2C2_SDA	PB11 (48)	AF I2C1	MS5607, LIS3MDL
SPI3_SCK	PB3 (89)	AF SPI3	Flash and LoRa
SPI3_MISO	PB4 (90)	AF SPI3	Flash and LoRa
SPI3_MOSI	PB5 (91)	AF SPI3	Flash and LoRa
CS_FLASH	PD6 (87)	GPIO output	W25Q128JVS
CS_LORA	PE4 (3)	GPIO output	RFM95W-915
RFM_RESET	PE5 (4)	GPIO output	RFM95W-915
RFM_DIO0	PE3 (2)	GPIO input EXTI	RFM95W-915
SDIO_D0	PC8 (65)	AF SDIO	microSD
SDIO_D1	PC9 (66)	AF SDIO	microSD
SDIO_D2	PC10 (78)	AF SDIO	microSD
SDIO_D3	PC11 (79)	AF SDIO	microSD
SDIO_CK	PC12 (80)	AF SDIO	microSD
SDIO_CMD	PD2 (83)	AF SDIO	microSD
GPS_TX_TO_MCU_RX	PA1 (24)	UART RX	MAX-M10S TXD
GPS_RX_FROM_MCU_TX	PA0 (23)	UART TX	MAX-M10S RXD
GPS_TIMEPULSE	PA2 (25)	GPIO input EXTI	MAX-M10S TIMEPULSE
GPS_RESET_N	TEST PAD	GPIO output	MAX-M10S reset
USB_DM	PA11 (70)	USB OTG FS	USB-C connector
USB_DP	PA12 (71)	USB OTG FS	USB-C connector
SWDIO	PA13 (72)	SWD	ST-Link
SWCLK	PA14 (76)	SWD	ST-Link
NRST	NRST (14)	Reset	Reset circuit and SWD

BOOT0	BOOT0 (94)	Boot mode	Boot switch
BATT_VSENSE	PC2 (17)	ADC input	Battery divider
MAIN_FIRE	PD14 (61)	GPIO output	Main ejection circuit
DROGUE_FIRE	PD11 (58)	GPIO output	Drogue ejection circuit
MAIN_ARM/CONTINUITY	PD15(62)	Sense input	Main continuity circuit
DROGUE_ARM/CONTINUITY	PD10(57)	Sense input	Drogue continuity circuit
MAIN_EMATCH_SENSE	PD13(60)	Sense input	Main op-amp sense
DROGUE_EMATCH_SENSE	PD9(56)	Sense input	Drogue op-amp sense
CAM_PWR_CNT	PE2 (1)	GPIO output	Camera power control
BUZZER	PD3 (84)	GPIO or timer PWM	Buzzer
LED_STATUS	PE1 (98)	GPIO output	Status LED
LED_FAULT	PB12 (51)	GPIO output	Fault LED

Required Startup Defaults

Output group	Required startup state	Reason
MAIN_FIRE	OFF	Prevent accidental main charge firing. GPIO has a pulldown
DROGUE_FIRE	OFF	Prevent accidental drogue charge firing. GPIO has a pulldown
CS lines	High	Prevent SPI bus contention.
RFM_RESET	Not held in reset unless driver requests it	Allows radio startup control.
CAM_PWR_EN	Default ON	Avoid random camera power cycling.
LEDs and buzzer	Off until firmware starts status task	Avoid confusing boot behavior.

Software Ownership Split

Area	Owner task
Board support package	Clock setup, GPIO init, DMA, timers, watchdog, fault reset reason.
Motion drivers	LSM6DS3TR-C, BMI270, ADXL375, data-ready, scaling to units.
Environmental drivers	MS5607 altitude, LIS3MDL heading reference data.
Power drivers	INA260, battery ADC, battery status.
Storage	microSD logging, W25Q flash black box, log file format.
Telemetry	RFM95W packet format, radio init, packet scheduling.
GPS	MAX-M10S parser, fix status, GPS time, location packet.
Recovery	Launch detect, apogee detect, drogue fire, main fire, continuity, lockouts.
State machine	Initializing, wait for launch, ascent, descent, drogue descent, main descent, landed, fault.
Test tools	Bench test firmware, fake sensor input, log replay, packet decoder.